

STAT405 Bayesian Statistics Final Project

EDA

```
library(ggplot2)
library(dplyr)

##
## Attaching package: 'dplyr'
##
## The following objects are masked from 'package:stats':
##
##   filter, lag
##
## The following objects are masked from 'package:base':
##
##   intersect, setdiff, setequal, union
library(corrplot)

## corrplot 0.95 loaded
library(car)

## Loading required package: carData
##
## Attaching package: 'car'
##
## The following object is masked from 'package:dplyr':
##
##   recode
source("simple.R")
source("simple_utils.R")

house_price <- read.csv("data/house-prices-advanced-regression-techniques/train.csv")

# check MSSubClass, it should be character
house_price$MSSubClass <- as.character(house_price$MSSubClass)

# Replace NA with explicit categories
house_price$Alley[is.na(house_price$Alley)] <- "NoAlley"
house_price$GarageType[is.na(house_price$GarageType)] <- "NoGarageType"
house_price$MasVnrType[is.na(house_price$MasVnrType)] <- "NoMasVnrType"
house_price$MasVnrArea[is.na(house_price$MasVnrArea)] <- 0
house_price$Fence[is.na(house_price$Fence)] <- "NoFence"
house_price$MiscFeature[is.na(house_price$MiscFeature)] <- "NoMiscFeature"

house_price$BsmtQual[is.na(house_price$BsmtQual)] <- "NoBsmt"
house_price$BsmtCond[is.na(house_price$BsmtCond)] <- "NoBsmt"
```

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house_price$BsmtExposure[is.na(house_price$BsmtExposure)] <- "NoBsmt"
house_price$BsmtFinType1[is.na(house_price$BsmtFinType1)] <- "NoBsmt"
house_price$BsmtFinType2[is.na(house_price$BsmtFinType2)] <- "NoBsmt"
house_price$FireplaceQu[is.na(house_price$FireplaceQu)] <- "NoFireplace"
house_price$GarageFinish[is.na(house_price$GarageFinish)] <- "NoGarage"
house_price$GarageQual[is.na(house_price$GarageQual)] <- "NoGarage"
house_price$GarageCond[is.na(house_price$GarageCond)] <- "NoGarage"
house_price$PoolQC[is.na(house_price$PoolQC)] <- "NoPool"

# Add indicator variables first
house_price$HasBsmt <- as.factor(ifelse(house_price$TotalBsmtSF > 0, 1, 0))
house_price$HasGarage <- as.factor(ifelse(house_price$GarageArea > 0, 1, 0))
house_price$HasFireplace <- as.factor(ifelse(house_price$Fireplaces > 0, 1, 0))
house_price$HasPool <- as.factor(ifelse(house_price$PoolArea > 0, 1, 0))
house_price$HasMasVnr <- as.factor(ifelse(house_price$MasVnrArea > 0, 1, 0))

# Convert to ordered factors
qual_levels <- c("Po", "Fa", "TA", "Gd", "Ex")
house_price$ExterQual <- factor(house_price$ExterQual,
                                levels = qual_levels, ordered = TRUE)
house_price$ExterCond <- factor(house_price$ExterCond,
                                levels = qual_levels, ordered = TRUE)
house_price$BsmtQual <- factor(house_price$BsmtQual,
                                levels = c("NoBsmt", "Po", "Fa", "TA", "Gd", "Ex"),
                                ordered = TRUE)
house_price$BsmtCond <- factor(house_price$BsmtCond,
                                levels = c("NoBsmt", "Po", "Fa", "TA", "Gd", "Ex"),
                                ordered = TRUE)
house_price$BsmtExposure <- factor(house_price$BsmtExposure,
                                    levels = c("NoBsmt", "No", "Mn", "Av", "Gd"),
                                    ordered = TRUE)
house_price$BsmtFinType1 <- factor(house_price$BsmtFinType1,
                                    levels = c("NoBsmt", "Unf", "LwQ", "Rec", "BLQ", "ALQ", "GLQ"),
                                    ordered = TRUE)
house_price$BsmtFinType2 <- factor(house_price$BsmtFinType2,
                                    levels = c("NoBsmt", "Unf", "LwQ", "Rec", "BLQ", "ALQ", "GLQ"),
                                    ordered = TRUE)
house_price$HeatingQC <- factor(house_price$HeatingQC,
                                levels = qual_levels, ordered = TRUE)
house_price$KitchenQual <- factor(house_price$KitchenQual,
                                levels = qual_levels, ordered = TRUE)
house_price$Functional <- factor(house_price$Functional,
                                levels = c("Sal", "Sev", "Maj2", "Maj1", "Mod", "Min2", "Min1", "Typ"),
                                ordered = TRUE)
house_price$FireplaceQu <- factor(house_price$FireplaceQu,
                                    levels = c("NoFireplace", "Po", "Fa", "TA", "Gd", "Ex"),
                                    ordered = TRUE)
house_price$GarageFinish <- factor(house_price$GarageFinish,
                                    levels = c("NoGarage", "Unf", "RFn", "Fin"),
                                    ordered = TRUE)
house_price$GarageQual <- factor(house_price$GarageQual,
                                    levels = c("NoGarage", "Po", "Fa", "TA", "Gd", "Ex"),
                                    ordered = TRUE)

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house_price$GarageCond <- factor(house_price$GarageCond,
                                levels = c("NoGarage", "Po", "Fa", "TA", "Gd", "Ex"),
                                ordered = TRUE)
house_price$PoolQC <- factor(house_price$PoolQC,
                             levels = c("NoPool", "Fa", "TA", "Gd", "Ex"),
                             ordered = TRUE)
house_price$LotShape <- factor(house_price$LotShape,
                               levels = c("IR3", "IR2", "IR1", "Reg"),
                               ordered = TRUE)
house_price$LandSlope <- factor(house_price$LandSlope,
                                levels = c("Sev", "Mod", "Gtl"),
                                ordered = TRUE)
house_price$PavedDrive <- factor(house_price$PavedDrive,
                                 levels = c("N", "P", "Y"),
                                 ordered = TRUE)

# Convert ordered factors to numeric scores
ordinal_to_numeric <- function(x) as.numeric(x) - 1

ord_vars <- c(
  "ExterQual", "ExterCond",
  "BsmtQual", "BsmtCond", "BsmtExposure", "BsmtFinType1", "BsmtFinType2",
  "HeatingQC", "KitchenQual", "Functional", "FireplaceQu",
  "GarageFinish", "GarageQual", "GarageCond",
  "PoolQC", "LotShape", "LandSlope", "PavedDrive"
)
house_price[ord_vars] <- lapply(house_price[ord_vars], ordinal_to_numeric)
house_price <- na.omit(house_price)

cat_vars <- names(house_price)[sapply(house_price, function(x) is.factor(x) || is.character(x))]
for (v in cat_vars) {
  cat("\n-----\n")
  cat("Variable:", v, "\n")
  print(table(house_price[[v]], useNA = "ifany"))
}

```

```

##
## -----
## Variable: MSSubClass
##
## 120 160 180 190 20 30 40 45 50 60 70 75 80 85 90
## 67 53 6 19 425 47 3 9 116 230 52 13 37 14 35
##
## -----
## Variable: MSZoning
##
## C (all) FV RH RL RM
## 8 57 10 876 175
##
## -----
## Variable: Street
##
## Grvl Pave
## 4 1122

```

```

##
## -----
## Variable: Alley
##
##      Grvl NoAlley      Pave
##      41      1048      37
##
## -----
## Variable: LandContour
##
##      Bnk  HLS   Low  Lvl
##      47   44   15 1020
##
## -----
## Variable: Utilities
##
## AllPub
##      1126
##
## -----
## Variable: LotConfig
##
##      Corner CulDSac      FR2      FR3  Inside
##      191      44      30      4      857
##
## -----
## Variable: Neighborhood
##
## Blmngtn Blueste  BrDale BrkSide ClearCr CollgCr Crawfor Edwards Gilbert  IDOTRR
##      14      2      15      42      12      124      41      71      49      27
## MeadowV Mitchel  NAmes NoRidge NPKvill NridgHt  NWAmes OldTown  Sawyer SawyerW
##      10      32      181      33      7      76      45      97      46      49
## Somerst StoneBr  SWISU  Timber Veenker
##      78      20      19      29      7
##
## -----
## Variable: Condition1
##
## Artery  Feedr  Norm  PosA  PosN  RRAe  RRAn  RRNe  RRNn
##      43      56      976      4      9      9      24      1      4
##
## -----
## Variable: Condition2
##
## Artery  Feedr  Norm  PosA  PosN  RRNn
##      2      5      1114      1      2      2
##
## -----
## Variable: BldgType
##
##      1Fam 2fmCon Duplex  Twnhs TwnhsE
##      945      20      35      35      91
##
## -----

```

```

## Variable: HouseStyle
##
## 1.5Fin 1.5Unf 1Story 2.5Fin 2.5Unf 2Story SFoyer SLvl
##    121    10    563        6        9    350    24    43
##
## -----
## Variable: RoofStyle
##
##    Flat    Gable Gambrel    Hip Mansard
##        6    870    10    234        6
##
## -----
## Variable: RoofMatl
##
## ClyTile CompShg Membran    Roll Tar&Grv WdShake WdShngl
##        1    1109        1        1        6        2        6
##
## -----
## Variable: Exterior1st
##
## AsbShng BrkComm BrkFace  CBlock CemntBd HdBoard ImStucc MetalSd Plywood  Stone
##        14        2    40        1    47    155        1    175    68        1
## Stucco VinylSd Wd Sdng WdShng
##        19    430    151    22
##
## -----
## Variable: Exterior2nd
##
## AsbShng AsphShn Brk Cmn BrkFace  CBlock CmentBd HdBoard ImStucc MetalSd  Other
##        15        2        7    21        1    47    142        9    172        1
## Plywood  Stone  Stucco VinylSd Wd Sdng Wd Shng
##        87        4    20    421    148    29
##
## -----
## Variable: MasVnrType
##
##        BrkCmn    BrkFace NoMasVnrType    None    Stone
##            9    333        6    659    119
##
## -----
## Variable: Foundation
##
## BrkTil CBlock PConc    Slab  Stone  Wood
##    122    453    528    15    6    2
##
## -----
## Variable: Heating
##
## GasA GasW Grav OthW Wall
## 1103  17    2    1    3
##
## -----
## Variable: CentralAir
##

```

```

##      N      Y
##    63 1063
##
## -----
## Variable: Electrical
##
## FuseA FuseF FuseP   Mix SBrkr
##    70    20    2    1  1033
##
## -----
## Variable: GarageType
##
## 2Types  Attchd Basment BuiltIn CarPort  Detchd
##      5    694    15    64    8    340
##
## -----
## Variable: Fence
##
##   GdPrv   GdWo   MnPrv   MnWw NoFence
##     47     39    119     10    911
##
## -----
## Variable: MiscFeature
##
## NoMiscFeature      Othr      Shed      TenC
##          1090          1          34          1
##
## -----
## Variable: SaleType
##
##   COD   Con ConLD ConLI ConLw   CWD   New   Oth   WD
##   32    2    6    3    4    4   119    1   955
##
## -----
## Variable: SaleCondition
##
## Abnorml AdjLand  Alloca  Family  Normal  Partial
##      71      1      9      18    905    122
##
## -----
## Variable: HasBsmt
##
##      0      1
##    24 1102
##
## -----
## Variable: HasGarage
##
##      0      1
##      0 1126
##
## -----
## Variable: HasFireplace
##

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```

##    0    1
## 533 593
##
## -----
## Variable: HasPool
##
##    0    1
## 1120   6
##
## -----
## Variable: HasMasVnr
##
##    0    1
## 664 462

# month as time seires
house_price$month_sin <- sin(2 * pi * house_price$MoSold / 12)
house_price$month_cos <- cos(2 * pi * house_price$MoSold / 12)
#
house_price$YearSinceRemodel <- house_price$YrSold - house_price$YearRemodAdd
house_price$YearSinceBuild <- house_price$YrSold - house_price$YearBuilt

house_price <- house_price |>
  select(-Id, -MoSold, -YrSold, -YearRemodAdd, -YearBuilt,
         # only one level
         -HasGarage, -Utilities,
         # too sparse
         -Street, -Condition2, -RoofMatl, -Heating, -HasPool, -MiscFeature)

# merge some level
house_price$Condition1 <- dplyr::case_when(
  house_price$Condition1 == "Norm" ~ "Norm",
  house_price$Condition1 %in% c("Artery", "Feedr") ~ "Road",
  house_price$Condition1 %in% c("RRAn", "RRAe", "RRNe", "RRNn") ~ "Railroad",
  house_price$Condition1 %in% c("PosA", "PosN") ~ "PositiveFeature",
  TRUE ~ "Other"
)
house_price$Condition1 <- factor(house_price$Condition1)
house_price$RoofStyle <- dplyr::case_when(
  house_price$RoofStyle %in% c("Gable", "Hip") ~ house_price$RoofStyle,
  TRUE ~ "Other"
)
house_price$RoofStyle <- factor(house_price$RoofStyle)
house_price$Exterior1st <- dplyr::case_when(
  house_price$Exterior1st %in% c("VinylSd", "MetalSd", "HdBoard", "Wd Sdng", "Plywood", "CemntBd") ~ house_price$Exterior1st,
  house_price$Exterior1st %in% c("BrkFace", "Stone", "Stucco", "WdShing", "AsbShng") ~ house_price$Exterior1st,
  TRUE ~ "Other"
)
house_price$Exterior1st <- factor(house_price$Exterior1st)
house_price$Exterior2nd <- dplyr::case_when(
  house_price$Exterior2nd %in% c("VinylSd", "MetalSd", "HdBoard", "Wd Sdng", "Plywood", "CmentBd", "Wd Shng") ~ house_price$Exterior2nd,
  house_price$Exterior2nd %in% c("BrkFace", "Stucco", "AsbShng") ~ house_price$Exterior2nd,
  TRUE ~ "Other"
)

```

```

house_price$Exterior2nd <- factor(house_price$Exterior2nd)
house_price$Electrical <- dplyr::case_when(
  house_price$Electrical == "SBrkr" ~ "SBrkr",
  house_price$Electrical %in% c("FuseA", "FuseF") ~ "Fuse",
  house_price$Electrical %in% c("FuseP", "Mix") ~ "PoorOrMixed",
  TRUE ~ "Other"
)
house_price$Electrical <- factor(house_price$Electrical)
house_price$GarageType <- dplyr::case_when(
  house_price$GarageType == "Attchd" ~ "Attached",
  house_price$GarageType == "Detchd" ~ "Detached",
  house_price$GarageType == "BuiltIn" ~ "BuiltIn",
  house_price$GarageType %in% c("Basment", "CarPort", "2Types") ~ "OtherGarage",
  TRUE ~ "OtherGarage"
)
house_price$GarageType <- factor(house_price$GarageType)
house_price$SaleType <- dplyr::case_when(
  house_price$SaleType == "WD" ~ "WarrantyDeed",
  house_price$SaleType == "New" ~ "NewHome",
  house_price$SaleType == "COD" ~ "CourtOrEstate",
  house_price$SaleType %in% c("CWD", "Con", "ConLD", "ConLI", "ConLw", "Oth") ~ "OtherSaleType",
  TRUE ~ "OtherSaleType"
)
house_price$SaleType <- factor(house_price$SaleType)
house_price$SaleCondition <- dplyr::case_when(
  house_price$SaleCondition == "Normal" ~ "Normal",
  house_price$SaleCondition == "Abnorml" ~ "Abnormal",
  house_price$SaleCondition == "Partial" ~ "Partial",
  house_price$SaleCondition %in% c("AdjLand", "Alloca", "Family") ~ "OtherCondition",
  TRUE ~ "OtherCondition"
)
house_price$SaleCondition <- factor(house_price$SaleCondition)

cat_vars <- names(house_price)[sapply(house_price, function(x) is.factor(x) || is.character(x))]
for (v in cat_vars) {
  cat("\n-----\n")
  cat("Variable:", v, "\n")
  print(table(house_price[[v]], useNA = "ifany"))
}

```

```

##
## -----
## Variable: MSSubClass
##
## 120 160 180 190 20 30 40 45 50 60 70 75 80 85 90
## 67 53 6 19 425 47 3 9 116 230 52 13 37 14 35
##
## -----
## Variable: MSZoning
##
## C (all) FV RH RL RM
## 8 57 10 876 175
##
## -----

```



```

## Variable: Alley
##
##      Grvl NoAlley      Pave
##      41      1048      37
##
## -----
## Variable: LandContour
##
##      Bnk  HLS  Low  Lvl
##      47   44   15 1020
##
## -----
## Variable: LotConfig
##
##      Corner CulDSac      FR2      FR3  Inside
##      191      44      30      4      857
##
## -----
## Variable: Neighborhood
##
##      Blmngtn Blueste  BrDale BrkSide ClearCr CollgCr Crawfor Edwards Gilbert  IDOTRR
##      14      2      15      42      12      124      41      71      49      27
##      MeadowV Mitchel  NAmes NoRidge NPKvill NridgHt  NWAmes OldTown  Sawyer SawyerW
##      10      32      181      33      7      76      45      97      46      49
##      Somerst StoneBr  SWISU  Timber Veenker
##      78      20      19      29      7
##
## -----
## Variable: Condition1
##
##      Norm PositiveFeature      Railroad      Road
##      976      13      38      99
##
## -----
## Variable: BldgType
##
##      1Fam 2fmCon Duplex  Twnhs TwnhsE
##      945      20      35      35      91
##
## -----
## Variable: HouseStyle
##
##      1.5Fin 1.5Unf 1Story 2.5Fin 2.5Unf 2Story SFoyer  SLvl
##      121      10      563      6      9      350      24      43
##
## -----
## Variable: RoofStyle
##
##      Gable  Hip Other
##      870      234      22
##
## -----
## Variable: Exterior1st
##

```

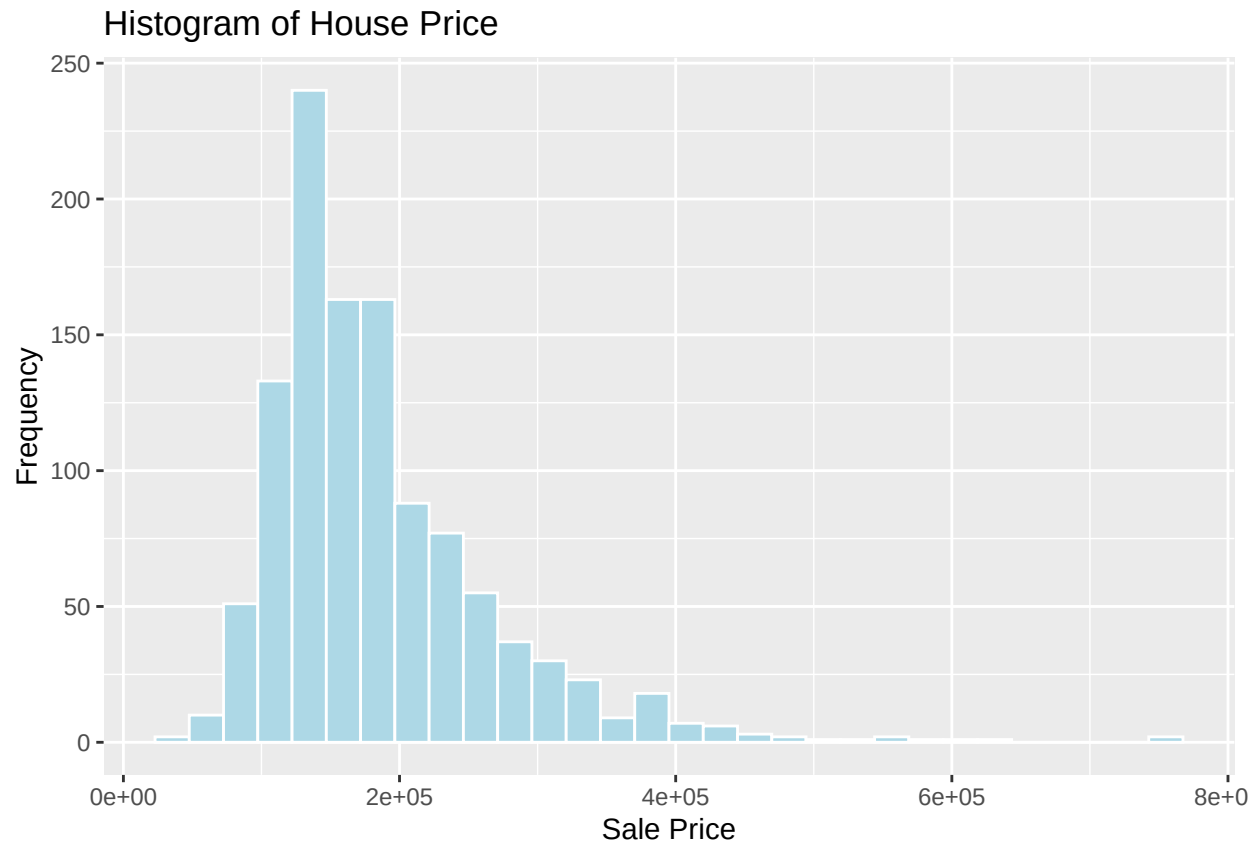
```

## AsbShng BrkFace CemntBd HdBoard MetalSd Other Plywood Stone Stucco VinylSd
##      14      40      47      155      175      4      68      1      19      430
## Wd Sdng WdShng
##      151      22
##
## -----
## Variable: Exterior2nd
##
## AsbShng BrkFace CemntBd HdBoard MetalSd Other Plywood Stucco VinylSd Wd Sdng
##      15      21      47      142      172      24      87      20      421      148
## Wd Shng
##      29
##
## -----
## Variable: MasVnrType
##
##      BrkCmn      BrkFace NoMasVnrType      None      Stone
##          9          333          6          659          119
##
## -----
## Variable: Foundation
##
## BrkTil CBlock PConc Slab Stone Wood
##      122      453      528      15      6      2
##
## -----
## Variable: CentralAir
##
##      N      Y
##      63 1063
##
## -----
## Variable: Electrical
##
##      Fuse PoorOrMixed      SBrkr
##          90          3          1033
##
## -----
## Variable: GarageType
##
##      Attached      BuiltIn      Detached OtherGarage
##          694          64          340          28
##
## -----
## Variable: Fence
##
##      GdPrv      GdWo      MnPrv      MnWw NoFence
##          47          39          119          10          911
##
## -----
## Variable: SaleType
##
## CourtOrEstate      NewHome OtherSaleType WarrantyDeed
##          32          119          20          955

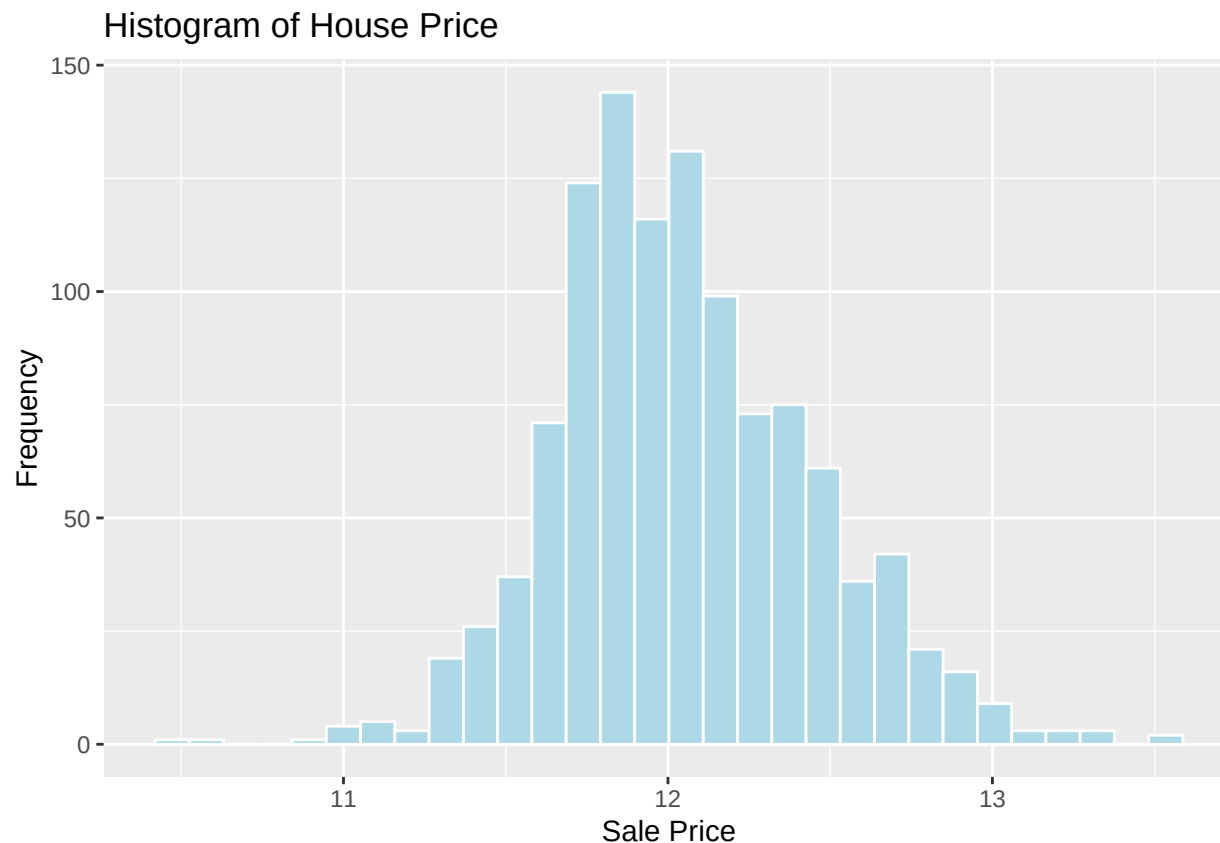
```

```
##
## -----
## Variable: SaleCondition
##
##      Abnormal      Normal OtherCondition      Partial
##           71           905           28           122
##
## -----
## Variable: HasBsmt
##
##      0      1
##    24 1102
##
## -----
## Variable: HasFireplace
##
##      0      1
##   533 593
##
## -----
## Variable: HasMasVnr
##
##      0      1
##   664 462

# check distribution of Y
ggplot(house_price, aes(x = SalePrice)) +
  geom_histogram(bins = 30, fill = "lightblue", color = "white") +
  labs(title = "Histogram of House Price",
       x = "Sale Price",
       y = "Frequency")
```



```
# Right skew -> log(SalePrice)
house_price$LogSalePrice = log(house_price$SalePrice)
ggplot(house_price, aes(x = LogSalePrice)) +
  geom_histogram(bins = 30, fill = "lightblue", color = "white") +
  labs(title = "Histogram of House Price",
       x = "Sale Price",
       y = "Frequency")
```



```
house_price <- house_price |> select(-SalePrice)
```

```
numeric_vars <- house_price[, sapply(house_price, is.numeric)]
cor_matrix <- cor(numeric_vars, use = "pairwise.complete.obs")
cor_with_log_price_abs <- abs(cor_matrix[, "LogSalePrice"])
cor_with_log_price_abs <- sort(
  cor_with_log_price_abs[names(cor_with_log_price_abs) != "LogSalePrice"],
  decreasing = TRUE
)
cor_with_log_price_abs
```

##	OverallQual	ExterQual	GrLivArea	KitchenQual
##	0.829026568	0.709547984	0.695477575	0.686360986
##	GarageCars	BsmtQual	GarageArea	TotalBsmtSF
##	0.665407947	0.647958389	0.627927567	0.618239836
##	FullBath	X1stFlrSF	YearSinceRemodel	YearSinceBuild
##	0.609465503	0.604950177	0.595291666	0.595220858
##	GarageFinish	GarageYrBlt	TotRmsAbvGrd	FireplaceQu
##	0.585711121	0.564066814	0.542626220	0.539809103
##	HeatingQC	Fireplaces	MasVnrArea	BsmtExposure
##	0.506222945	0.480893101	0.440431885	0.383174268
##	OpenPorchSF	BsmtFinSF1	LotFrontage	WoodDeckSF
##	0.373178823	0.372763649	0.354562115	0.346075611
##	BsmtFinType1	LotShape	X2ndFlrSF	HalfBath
##	0.335161548	0.332308393	0.298185483	0.298130317
##	LotArea	BsmtCond	BsmtFullBath	PavedDrive
##	0.281460581	0.246754055	0.243844335	0.232468012

```
##      BsmtUnfSF      BedroomAbvGr      GarageQual      EnclosedPorch
##      0.229931247      0.196913859      0.188022328      0.177411889
##      GarageCond      KitchenAbvGr      Functional      ScreenPorch
##      0.173775411      0.157999818      0.120560126      0.118121121
##      OverallCond      PoolQC      month_sin      PoolArea
##      0.093206171      0.077044913      0.067270916      0.065771781
##      month_cos      X3SsnPorch      MiscVal      BsmtHalfBath
##      0.046253492      0.037988471      0.035300817      0.032790298
##      LandSlope      BsmtFinSF2      BsmtFinType2      LowQualFinSF
##      0.026497431      0.019182272      0.018075556      0.013911951
##      ExterCond
##      0.003460266
```

```
drop_vars <- names(cor_with_log_price_abs[cor_with_log_price_abs < 0.3])
house_price <- house_price |> select(-all_of(drop_vars))
```

```
# drop high cor features
numeric_vars2 <- house_price[, sapply(house_price, is.numeric)]
numeric_vars2 <- numeric_vars2[, colnames(numeric_vars2) != "LogSalePrice"]
selected_cor_matrix <- cor(numeric_vars2, use = "pairwise.complete.obs")
high_cor_pairs <- which(
  abs(selected_cor_matrix) > 0.7 & upper.tri(selected_cor_matrix),
  arr.ind = TRUE
)
high_cor_df <- data.frame(
  var1 = rownames(selected_cor_matrix)[high_cor_pairs[, 1]],
  var2 = colnames(selected_cor_matrix)[high_cor_pairs[, 2]],
  cor = selected_cor_matrix[high_cor_pairs]
)
high_cor_df
```

```
##      var1      var2      cor
## 1 OverallQual      ExterQual 0.7498563
## 2 TotalBsmtSF      X1stFlrSF 0.8366715
## 3      ExterQual      KitchenQual 0.7324666
## 4      GrLivArea      TotRmsAbvGrd 0.8239077
## 5      Fireplaces      FireplaceQu 0.8677742
## 6      GarageCars      GarageArea 0.8400285
## 7      GarageYrBlt      YearSinceBuild -0.8230434
```

```
drop_vars_highcor <- c()

for (i in 1:nrow(high_cor_df)) {
  v1 <- high_cor_df$var1[i]
  v2 <- high_cor_df$var2[i]

  c1 <- cor_with_log_price_abs[v1]
  c2 <- cor_with_log_price_abs[v2]

  if (c1 < c2) {
    drop_vars_highcor <- c(drop_vars_highcor, v1)
  } else {
    drop_vars_highcor <- c(drop_vars_highcor, v2)
  }
}
```

```

drop_vars_highcor <- unique(drop_vars_highcor)
house_price <- house_price |> select(-all_of(drop_vars_highcor))

# check categorical features using ANOVA
# convert character to factor
house_price[] <- lapply(house_price, function(x) {
  if (is.character(x)) as.factor(x) else x
})
# select factor variables only
cat_features <- house_price[, sapply(house_price, is.factor)]

cat_features <- cat_features[, colnames(cat_features) != "LogSalePrice", drop = FALSE]
cat_pvalues_list <- lapply(names(cat_features), function(var) {
  fit <- aov(as.formula(paste("LogSalePrice ~", var)), data = house_price)
  p_val <- summary(fit)[[1]][["Pr(>F)"]][1]

  level_counts <- table(house_price[[var]])

  data.frame(
    variable = var,
    p_value = p_val,
    n_levels = nlevels(cat_features[[var]]),
    min_count = min(level_counts),
    stringsAsFactors = FALSE
  )
})
cat_pvalues <- do.call(rbind, cat_pvalues_list)
cat_pvalues <- cat_pvalues[order(cat_pvalues$p_value), ]
cat_pvalues

```

##	variable	p_value	n_levels	min_count
## 6	Neighborhood	2.176243e-189	25	2
## 14	Foundation	3.579833e-96	6	2
## 1	MSSubClass	5.482285e-83	15	3
## 17	GarageType	7.951334e-79	4	28
## 22	HasFireplace	1.564175e-72	2	533
## 13	MasVnrType	2.839508e-56	5	6
## 11	Exterior1st	6.552983e-53	12	1
## 12	Exterior2nd	1.637554e-50	11	15
## 2	MSZoning	6.295085e-46	5	8
## 23	HasMasVnr	6.464496e-43	2	462
## 20	SaleCondition	1.018836e-39	4	28
## 19	SaleType	8.419164e-38	4	20
## 15	CentralAir	1.206114e-25	2	63
## 16	Electrical	2.542585e-24	3	3
## 9	HouseStyle	1.238701e-22	8	6
## 10	RoofStyle	8.083170e-13	3	22
## 18	Fence	1.146946e-10	5	10
## 8	BldgType	1.000710e-09	5	20
## 7	Condition1	8.022774e-09	4	13
## 5	LotConfig	2.263226e-08	5	4
## 21	HasBsmt	2.988616e-08	2	24
## 3	Alley	6.203825e-08	3	37

```
## 4      LandContour 5.782590e-07      4      15
```

```
#full anova model
```

```
full_model <- lm(LogSalePrice ~ ., data = house_price)
alias(full_model)
```

```
## Model :
```

```
## LogSalePrice ~ MSSubClass + MSZoning + LotFrontage + Alley +
##   LotShape + LandContour + LotConfig + Neighborhood + Condition1 +
##   BldgType + HouseStyle + OverallQual + RoofStyle + Exterior1st +
##   Exterior2nd + MasVnrType + MasVnrArea + Foundation + BsmtQual +
##   BsmtExposure + BsmtFinType1 + BsmtFinSF1 + TotalBsmtSF +
##   HeatingQC + CentralAir + Electrical + GrLivArea + FullBath +
##   FireplaceQu + GarageType + GarageFinish + GarageCars + WoodDeckSF +
##   OpenPorchSF + Fence + SaleType + SaleCondition + HasBsmt +
##   HasFireplace + HasMasVnr + YearSinceRemodel + YearSinceBuild
##
```

```
## Complete :
```

```
##           (Intercept) MSSubClass160 MSSubClass180 MSSubClass190
## BldgTypeDuplex      0           0           0           0
## BldgTypeTwnhsE      1           0           0          -1
##           MSSubClass20 MSSubClass30 MSSubClass40 MSSubClass45 MSSubClass50
## BldgTypeDuplex      0           0           0           0           0
## BldgTypeTwnhsE     -1          -1          -1          -1          -1
##           MSSubClass60 MSSubClass70 MSSubClass75 MSSubClass80 MSSubClass85
## BldgTypeDuplex      0           0           0           0           0
## BldgTypeTwnhsE     -1          -1          -1          -1          -1
##           MSSubClass90 MSZoningFV MSZoningRH MSZoningRL MSZoningRM
## BldgTypeDuplex      1           0           0           0           0
## BldgTypeTwnhsE     -1           0           0           0           0
##           LotFrontage AlleyNoAlley AlleyPave LotShape LandContourHLS
## BldgTypeDuplex      0           0           0           0           0
## BldgTypeTwnhsE      0           0           0           0           0
##           LandContourLow LandContourLvl LotConfigCulDSac LotConfigFR2
## BldgTypeDuplex      0           0           0           0
## BldgTypeTwnhsE      0           0           0           0
##           LotConfigFR3 LotConfigInside NeighborhoodBlueste
## BldgTypeDuplex      0           0           0
## BldgTypeTwnhsE      0           0           0
##           NeighborhoodBrDale NeighborhoodBrkSide NeighborhoodClearCr
## BldgTypeDuplex      0           0           0
## BldgTypeTwnhsE      0           0           0
##           NeighborhoodCollgCr NeighborhoodCrawfor NeighborhoodEdwards
## BldgTypeDuplex      0           0           0
## BldgTypeTwnhsE      0           0           0
##           NeighborhoodGilbert NeighborhoodIDOTRR NeighborhoodMeadowV
## BldgTypeDuplex      0           0           0
## BldgTypeTwnhsE      0           0           0
##           NeighborhoodMitchel NeighborhoodNames NeighborhoodNoRidge
## BldgTypeDuplex      0           0           0
## BldgTypeTwnhsE      0           0           0
##           NeighborhoodNPkVill NeighborhoodNridgHt NeighborhoodNWAmes
## BldgTypeDuplex      0           0           0
## BldgTypeTwnhsE      0           0           0
##           NeighborhoodOldTown NeighborhoodSawyer NeighborhoodSawyerW
```


##	BldgTypeDuplex	0	0	0
##	BldgTypeTwnhsE	0	0	0
##		NeighborhoodSomerst	NeighborhoodStoneBr	NeighborhoodSWISU
##	BldgTypeDuplex	0	0	0
##	BldgTypeTwnhsE	0	0	0
##		NeighborhoodTimber	NeighborhoodVeenker	Condition1PositiveFeature
##	BldgTypeDuplex	0	0	0
##	BldgTypeTwnhsE	0	0	0
##		Condition1Railroad	Condition1Road	BldgType2fmCon BldgTypeTwnhs
##	BldgTypeDuplex	0	0	0
##	BldgTypeTwnhsE	0	0	-1
##		HouseStyle1.5Unf	HouseStyle1Story	HouseStyle2.5Fin
##	BldgTypeDuplex	0	0	0
##	BldgTypeTwnhsE	0	0	0
##		HouseStyle2.5Unf	HouseStyle2Story	HouseStyleSFoyer
##	BldgTypeDuplex	0	0	0
##	BldgTypeTwnhsE	0	0	0
##		HouseStyleSLvl	OverallQual	RoofStyleHip RoofStyleOther
##	BldgTypeDuplex	0	0	0
##	BldgTypeTwnhsE	0	0	0
##		Exterior1stBrkFace	Exterior1stCemntBd	Exterior1stHdBoard
##	BldgTypeDuplex	0	0	0
##	BldgTypeTwnhsE	0	0	0
##		Exterior1stMetalSd	Exterior1stOther	Exterior1stPlywood
##	BldgTypeDuplex	0	0	0
##	BldgTypeTwnhsE	0	0	0
##		Exterior1stStone	Exterior1stStucco	Exterior1stVinylSd
##	BldgTypeDuplex	0	0	0
##	BldgTypeTwnhsE	0	0	0
##		Exterior1stWd Sdng	Exterior1stWdShing	Exterior2ndBrkFace
##	BldgTypeDuplex	0	0	0
##	BldgTypeTwnhsE	0	0	0
##		Exterior2ndCmentBd	Exterior2ndHdBoard	Exterior2ndMetalSd
##	BldgTypeDuplex	0	0	0
##	BldgTypeTwnhsE	0	0	0
##		Exterior2ndOther	Exterior2ndPlywood	Exterior2ndStucco
##	BldgTypeDuplex	0	0	0
##	BldgTypeTwnhsE	0	0	0
##		Exterior2ndVinylSd	Exterior2ndWd Sdng	Exterior2ndWd Shng
##	BldgTypeDuplex	0	0	0
##	BldgTypeTwnhsE	0	0	0
##		MasVnrTypeBrkFace	MasVnrTypeNoMasVnrType	MasVnrTypeNone
##	BldgTypeDuplex	0	0	0
##	BldgTypeTwnhsE	0	0	0
##		MasVnrTypeStone	MasVnrArea	FoundationCBlock FoundationPConc
##	BldgTypeDuplex	0	0	0
##	BldgTypeTwnhsE	0	0	0
##		FoundationSlab	FoundationStone	FoundationWood BsmtQual
##	BldgTypeDuplex	0	0	0
##	BldgTypeTwnhsE	0	0	0
##		BsmtExposure	BsmtFinType1	BsmtFinSF1 TotalBsmtSF HeatingQC
##	BldgTypeDuplex	0	0	0
##	BldgTypeTwnhsE	0	0	0
##		CentralAirY	ElectricalPoorOrMixed	ElectricalSBrkr GrLivArea

```

## BldgTypeDuplex 0 0 0 0
## BldgTypeTwnhsE 0 0 0 0
## FullBath FireplaceQu GarageTypeBuiltIn GarageTypeDetached
## BldgTypeDuplex 0 0 0 0
## BldgTypeTwnhsE 0 0 0 0
## GarageTypeOtherGarage GarageFinish GarageCars WoodDeckSF
## BldgTypeDuplex 0 0 0 0
## BldgTypeTwnhsE 0 0 0 0
## OpenPorchSF FenceGdWo FenceMnPrv FenceMnWw FenceNoFence
## BldgTypeDuplex 0 0 0 0
## BldgTypeTwnhsE 0 0 0 0
## SaleTypeNewHome SaleTypeOtherSaleType SaleTypeWarrantyDeed
## BldgTypeDuplex 0 0 0
## BldgTypeTwnhsE 0 0 0
## SaleConditionNormal SaleConditionOtherCondition
## BldgTypeDuplex 0 0
## BldgTypeTwnhsE 0 0
## SaleConditionPartial HasBsmt1 HasFireplace1 HasMasVnr1
## BldgTypeDuplex 0 0 0 0
## BldgTypeTwnhsE 0 0 0 0
## YearSinceRemodel YearSinceBuild
## BldgTypeDuplex 0 0
## BldgTypeTwnhsE 0 0

```

```

house_price <- house_price|> select(-BldgType)
reduced_model <- lm(LogSalePrice ~ ., data = house_price)
anova(reduced_model)

```

```
## Analysis of Variance Table
```

```
##
```

```
## Response: LogSalePrice
```

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
## MSSubClass	14	56.743	4.0531	250.7691	< 2.2e-16 ***
## MSZoning	4	8.791	2.1977	135.9720	< 2.2e-16 ***
## LotFrontage	1	8.251	8.2508	510.4857	< 2.2e-16 ***
## Alley	2	0.199	0.0995	6.1545	0.0022057 **
## LotShape	1	3.932	3.9321	243.2857	< 2.2e-16 ***
## LandContour	3	1.410	0.4701	29.0829	< 2.2e-16 ***
## LotConfig	4	2.403	0.6007	37.1666	< 2.2e-16 ***
## Neighborhood	24	42.587	1.7745	109.7884	< 2.2e-16 ***
## Condition1	3	0.326	0.1086	6.7221	0.0001718 ***
## HouseStyle	7	0.684	0.0978	6.0491	6.291e-07 ***
## OverallQual	1	19.077	19.0769	1180.3161	< 2.2e-16 ***
## RoofStyle	2	0.210	0.1052	6.5094	0.0015537 **
## Exterior1st	11	1.541	0.1401	8.6679	8.642e-15 ***
## Exterior2nd	10	0.362	0.0362	2.2422	0.0138227 *
## MasVnrType	4	0.491	0.1226	7.5884	5.059e-06 ***
## MasVnrArea	1	0.261	0.2613	16.1669	6.239e-05 ***
## Foundation	5	0.700	0.1400	8.6615	4.766e-08 ***
## BsmtQual	1	1.185	1.1847	73.2998	< 2.2e-16 ***
## BsmtExposure	1	0.614	0.6138	37.9755	1.040e-09 ***
## BsmtFinType1	1	1.208	1.2079	74.7324	< 2.2e-16 ***
## BsmtFinSF1	1	0.298	0.2981	18.4467	1.919e-05 ***
## TotalBsmtSF	1	0.973	0.9729	60.1964	2.123e-14 ***
## HeatingQC	1	0.657	0.6568	40.6362	2.803e-10 ***

```
## CentralAir          1  0.600  0.6001  37.1298 1.579e-09 ***
## Electrical          2  0.307  0.1533   9.4821 8.333e-05 ***
## GrLivArea           1  4.694  4.6938 290.4135 < 2.2e-16 ***
## FullBath            1  0.138  0.1384   8.5654 0.0035042 **
## FireplaceQu         1  0.357  0.3571  22.0958 2.959e-06 ***
## GarageType          3  0.031  0.0104   0.6417 0.5882745
## GarageFinish        1  0.132  0.1319   8.1607 0.0043699 **
## GarageCars          1  1.238  1.2378  76.5836 < 2.2e-16 ***
## WoodDeckSF          1  0.077  0.0767   4.7444 0.0296281 *
## OpenPorchSF         1  0.123  0.1233   7.6314 0.0058420 **
## Fence               4  0.078  0.0195   1.2041 0.3074000
## SaleType            3  0.141  0.0469   2.9029 0.0339361 *
## SaleCondition        3  0.295  0.0984   6.0867 0.0004188 ***
## HasBsmt             1  0.004  0.0036   0.2254 0.6350810
## HasFireplace        1  0.012  0.0116   0.7181 0.3969574
## HasMasVnr           1  0.014  0.0144   0.8913 0.3453424
## YearSinceRemodel     1  0.292  0.2924  18.0941 2.301e-05 ***
## YearSinceBuild       1  0.000  0.0005   0.0306 0.8611011
## Residuals          994 16.066  0.0162
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
house_price <- house_price |>
  select(
    -GarageType,
    -Fence,
    -HasBsmt,
    -HasFireplace,
    -HasMasVnr,
    -YearSinceBuild,
    # *
    -Exterior2nd,
    -WoodDeckSF,
    -SaleType,
    # **
    -Alley,
    -RoofStyle,
    -FullBath,
    -GarageFinish,
    -OpenPorchSF
  )
full_model <- lm(LogSalePrice ~ ., data = house_price)
vif(full_model)
```

```
##              GVIF Df GVIF^(1/(2*Df))
## MSSubClass    2.621537e+07 14      1.840553
## MSZoning      3.798069e+01  4      1.575598
## LotFrontage   2.458219e+00  1      1.567871
## LotShape      1.587450e+00  1      1.259940
## LandContour   2.189240e+00  3      1.139503
## LotConfig     1.851199e+00  4      1.080020
## Neighborhood  2.328406e+04 24      1.233049
## Condition1    1.816462e+00  3      1.104598
## HouseStyle     8.716605e+05  7      2.656504
## OverallQual    4.526626e+00  1      2.127587
```

```
## Exterior1st      2.467700e+01 11      1.156874
## MasVnrType       4.452109e+00  4      1.205232
## MasVnrArea       2.614379e+00  1      1.616904
## Foundation       2.146147e+01  5      1.358833
## BsmtQual         4.158583e+00  1      2.039260
## BsmtExposure     1.908716e+00  1      1.381563
## BsmtFinType1     2.673976e+00  1      1.635230
## BsmtFinSF1       3.093953e+00  1      1.758964
## TotalBsmtSF      5.730376e+00  1      2.393820
## HeatingQC        2.140652e+00  1      1.463097
## CentralAir       1.693709e+00  1      1.301426
## Electrical       1.730547e+00  2      1.146954
## GrLivArea        6.685004e+00  1      2.585537
## FireplaceQu      1.918807e+00  1      1.385210
## GarageCars       2.430409e+00  1      1.558977
## SaleCondition    2.009979e+00  3      1.123394
## YearSinceRemodel 2.612821e+00  1      1.616422
```

```
str(house_price)
```

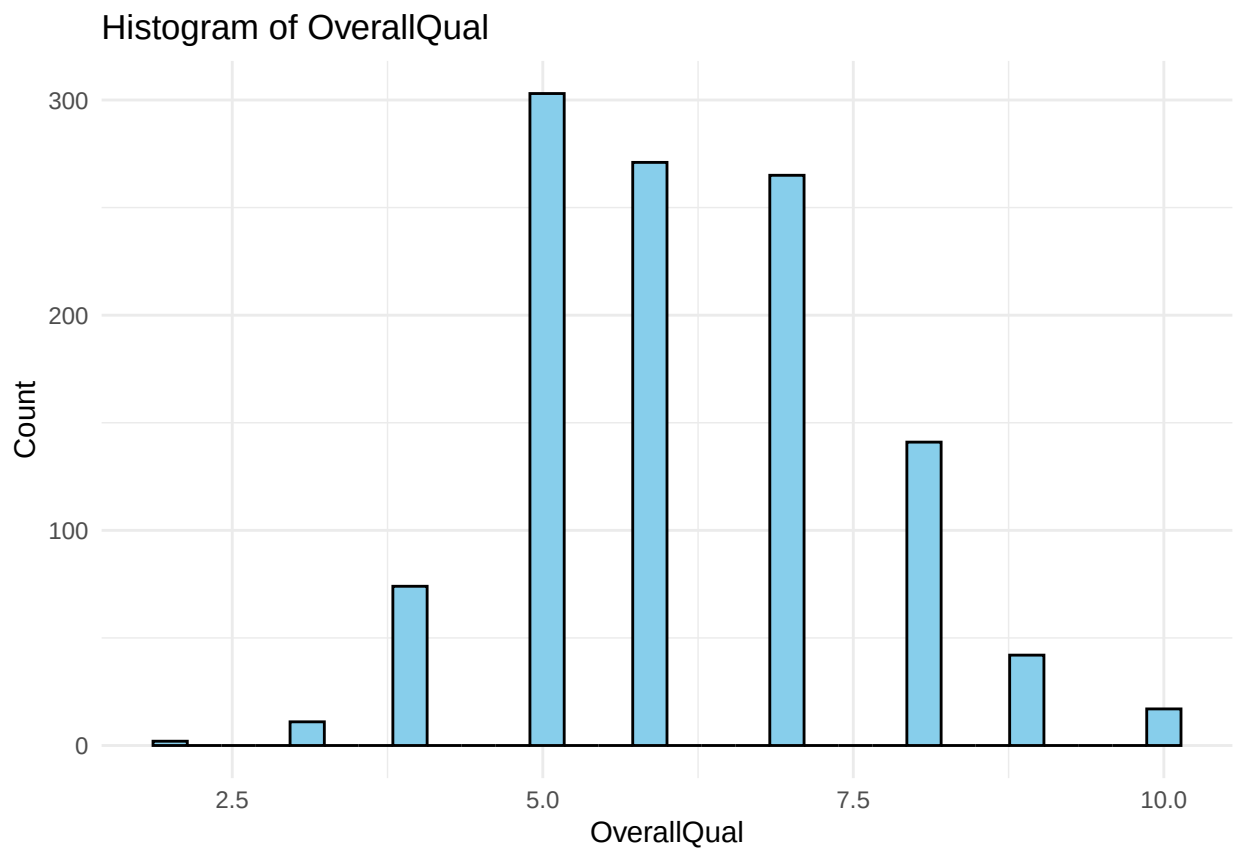
```
## 'data.frame':    1126 obs. of  28 variables:
## $ MSSubClass      : Factor w/ 15 levels "120","160","180",...: 10 5 10 11 10 9 5 9 4 5 ...
## $ MSZoning        : Factor w/ 5 levels "C (all)","FV",...: 4 4 4 4 4 4 5 4 4 ...
## $ LotFrontage     : int   65 80 68 60 84 85 75 51 50 70 ...
## $ LotShape        : num   3 3 2 2 2 2 3 3 3 3 ...
## $ LandContour     : Factor w/ 4 levels "Bnk","HLS","Low",...: 4 4 4 4 4 4 4 4 4 ...
## $ LotConfig       : Factor w/ 5 levels "Corner","CulDSac",...: 5 3 5 1 3 5 5 5 1 5 ...
## $ Neighborhood    : Factor w/ 25 levels "Blmngtn","Blueste",...: 6 25 6 7 14 12 21 18 4 19 ...
## $ Condition1      : Factor w/ 4 levels "Norm","PositiveFeature",...: 1 4 1 1 1 1 1 4 4 1 ...
## $ HouseStyle      : Factor w/ 8 levels "1.5Fin","1.5Unf",...: 6 3 6 6 6 1 3 1 2 3 ...
## $ OverallQual     : int   7 6 7 7 8 5 8 7 5 5 ...
## $ Exterior1st     : Factor w/ 12 levels "AsbShng","BrkFace",...: 10 5 10 11 10 10 10 2 5 4 ...
## $ MasVnrType      : Factor w/ 5 levels "BrkCmn","BrkFace",...: 2 4 2 4 2 4 5 4 4 4 ...
## $ MasVnrArea      : num   196 0 162 0 350 0 186 0 0 0 ...
## $ Foundation      : Factor w/ 6 levels "BrkTil","CBlock",...: 3 2 3 1 3 6 3 1 1 2 ...
## $ BsmtQual        : num   4 4 4 3 4 4 5 3 3 3 ...
## $ BsmtExposure    : num   1 4 2 1 3 1 3 1 1 1 ...
## $ BsmtFinType1    : num   6 5 6 5 6 6 6 1 6 3 ...
## $ BsmtFinSF1      : int   706 978 486 216 655 732 1369 0 851 906 ...
## $ TotalBsmtSF     : int   856 1262 920 756 1145 796 1686 952 991 1040 ...
## $ HeatingQC       : num   4 4 4 3 4 4 4 3 4 4 ...
## $ CentralAir      : Factor w/ 2 levels "N","Y": 2 2 2 2 2 2 2 2 2 ...
## $ Electrical      : Factor w/ 3 levels "Fuse","PoorOrMixed",...: 3 3 3 3 3 3 3 1 3 3 ...
## $ GrLivArea       : int   1710 1262 1786 1717 2198 1362 1694 1774 1077 1040 ...
## $ FireplaceQu     : num   0 3 3 4 3 0 4 3 3 0 ...
## $ GarageCars      : int   2 2 2 3 3 2 2 2 1 1 ...
## $ SaleCondition   : Factor w/ 4 levels "Abnormal","Normal",...: 2 2 2 1 2 2 2 1 2 2 ...
## $ YearSinceRemodel: int    5 31 6 36 8 14 2 58 58 43 ...
## $ LogSalePrice    : num   12.2 12.1 12.3 11.8 12.4 ...
## - attr(*, "na.action")= 'omit' Named int [1:334] 8 13 15 17 25 32 40 43 44 49 ...
## ..- attr(*, "names")= chr [1:334] "8" "13" "15" "17" ...
```

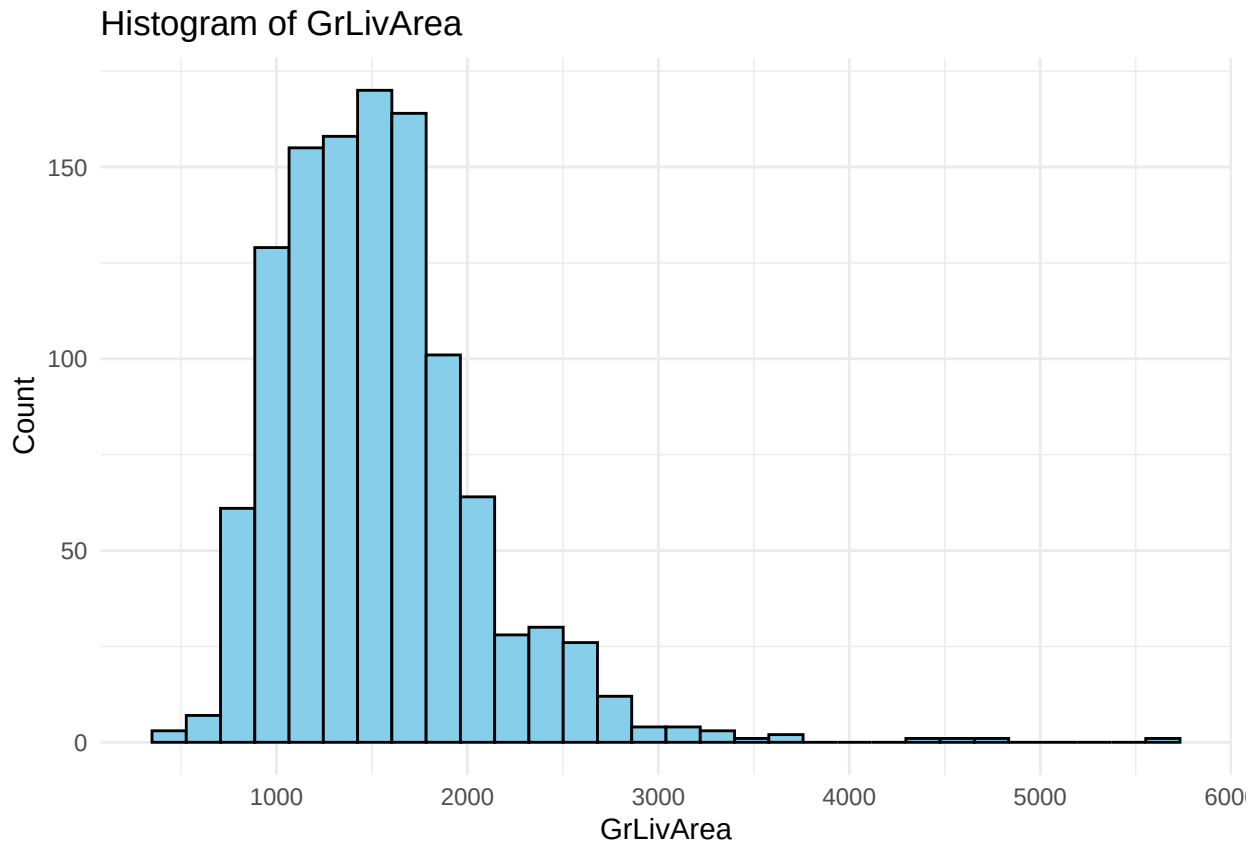
Model

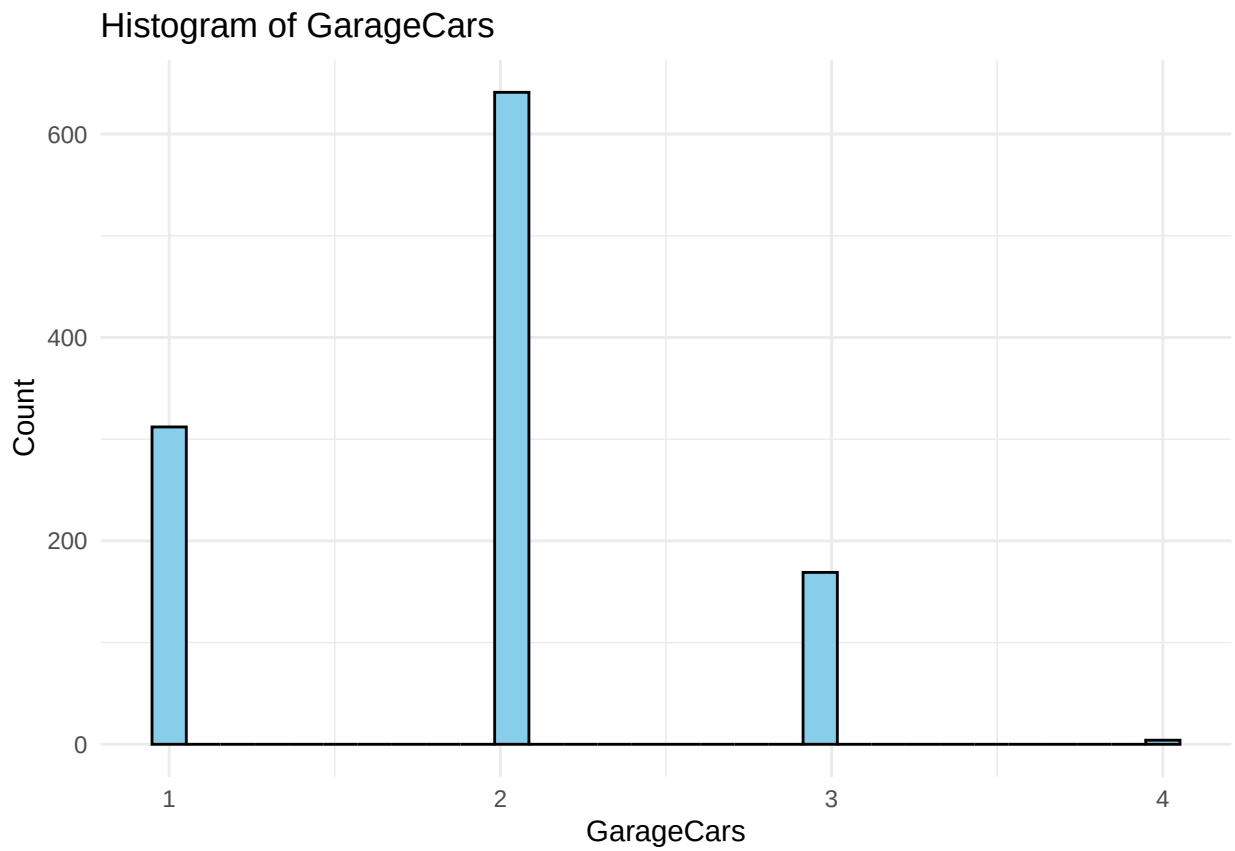
native model(only consider some numeric variable)

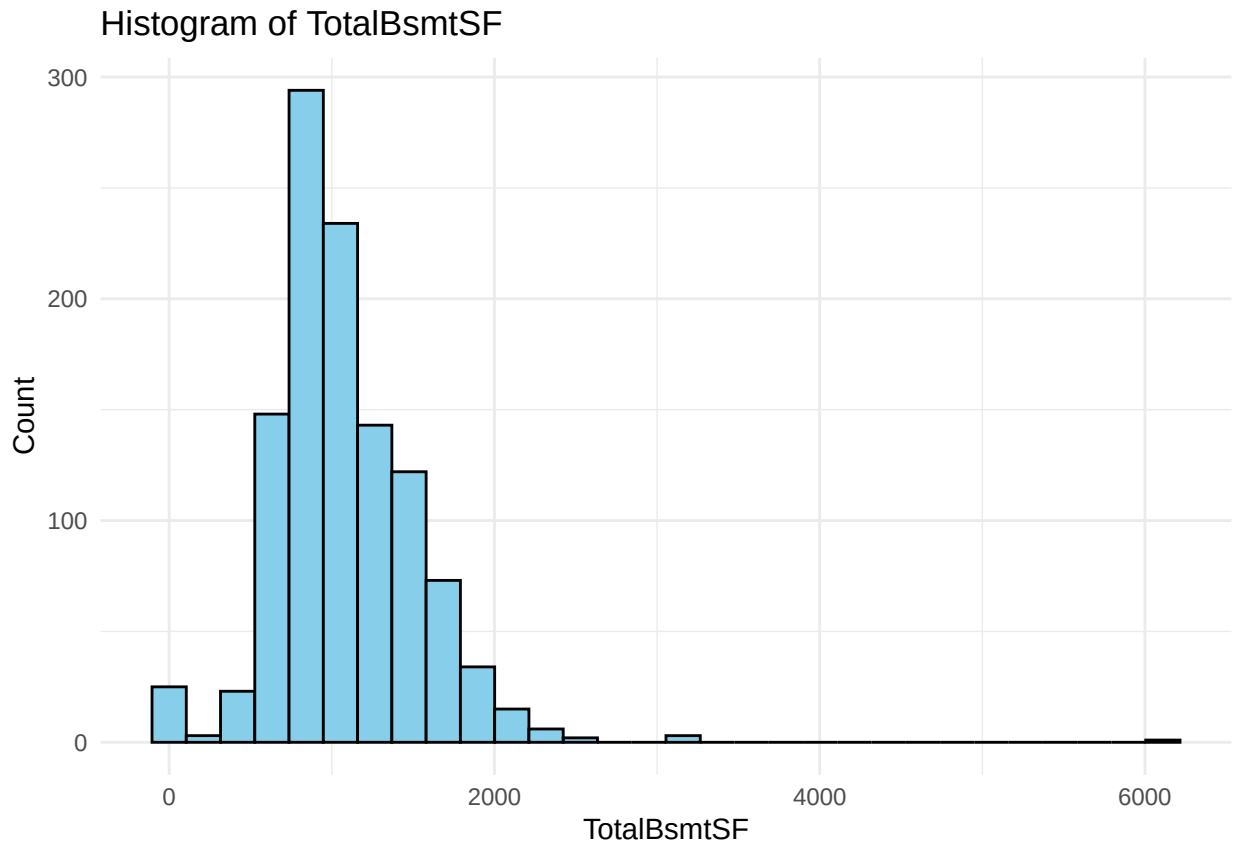
some reason:

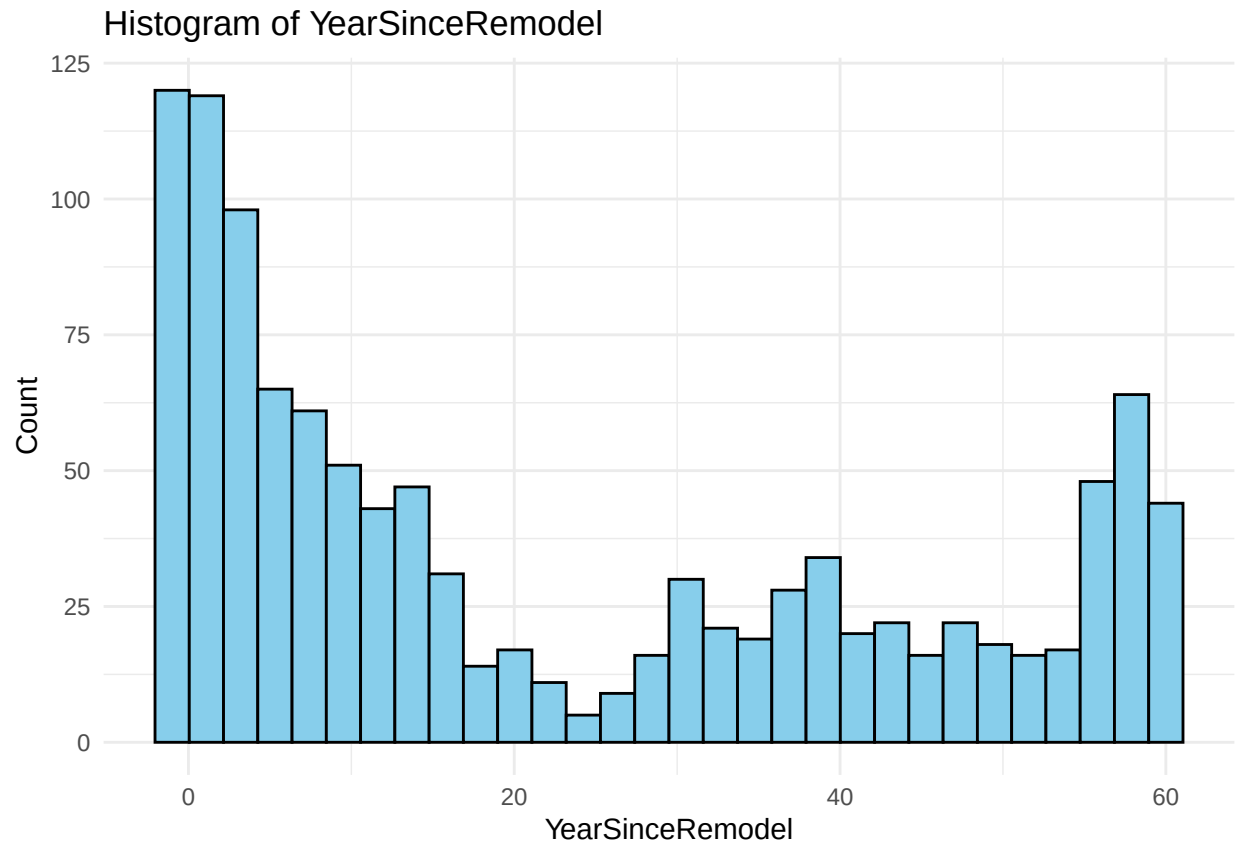
```
vars <- c(
  "OverallQual",
  "GrLivArea",
  "GarageCars",
  "TotalBsmtSF",
  "YearSinceRemodel"
)
for (v in vars) {
  p <- ggplot(house_price, aes(x = .data[[v]])) +
    geom_histogram(bins = 30, fill = "skyblue", color = "black") +
    labs(
      title = paste("Histogram of", v),
      x = v,
      y = "Count"
    ) +
    theme_minimal()
  print(p)
}
```











```
library(dplyr)
library(ggplot2)

# Since some predictors are right-skewed, we do feature transformation
house_price_naive <- house_price |>
  mutate(
    LogGrLivArea = log(GrLivArea),
  )

vars <- c(
  "OverallQual",
  "LogGrLivArea",
  "GarageCars",
  "TotalBsmntSF",
  "YearSinceRemodel"
)

for (v in vars) {
  p <- ggplot(house_price_naive, aes(x = .data[[v]], y = LogSalePrice)) +
    geom_point(alpha = 0.4) +
    geom_smooth(method = "lm", se = TRUE) +
    labs(
      title = paste("LogSalePrice vs", v),
      x = v,
      y = "LogSalePrice"
    ) +

```

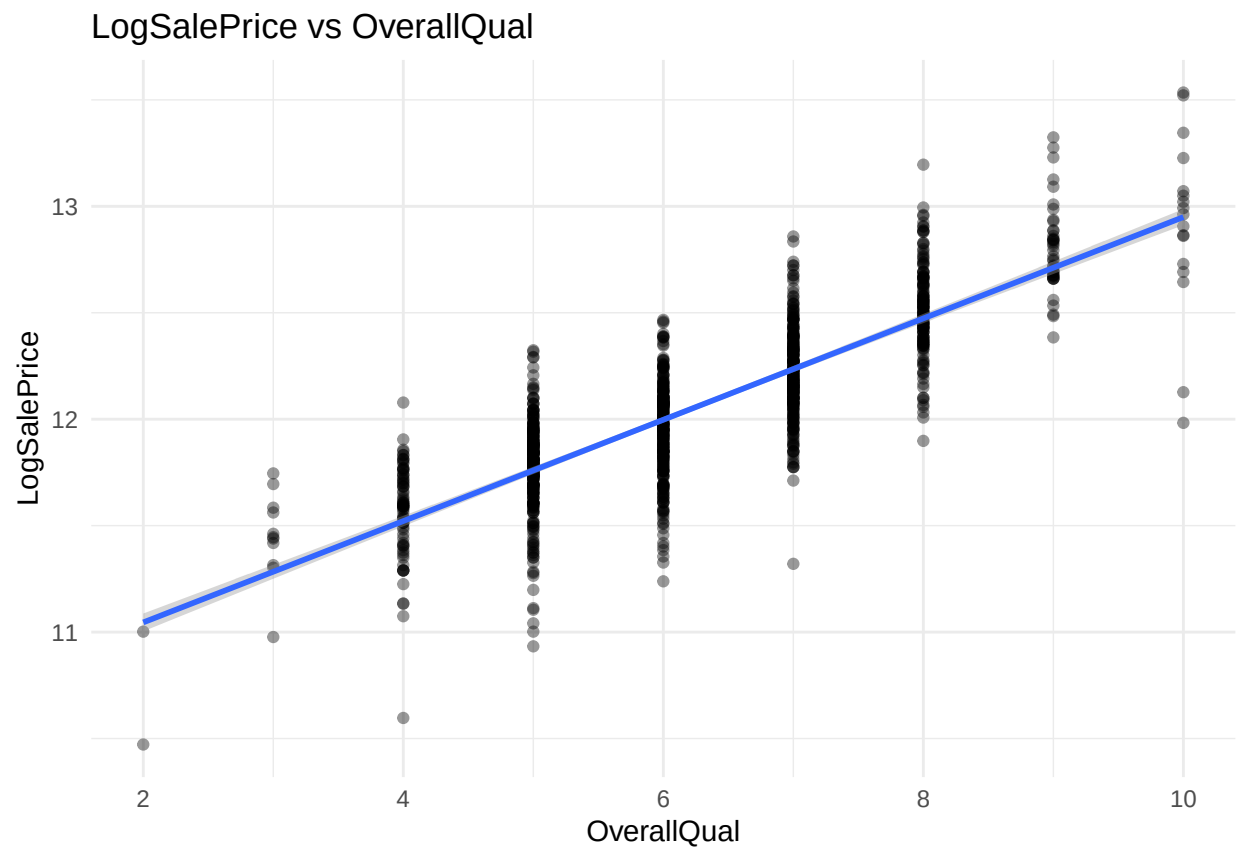
```

theme_minimal()

print(p)
}

```

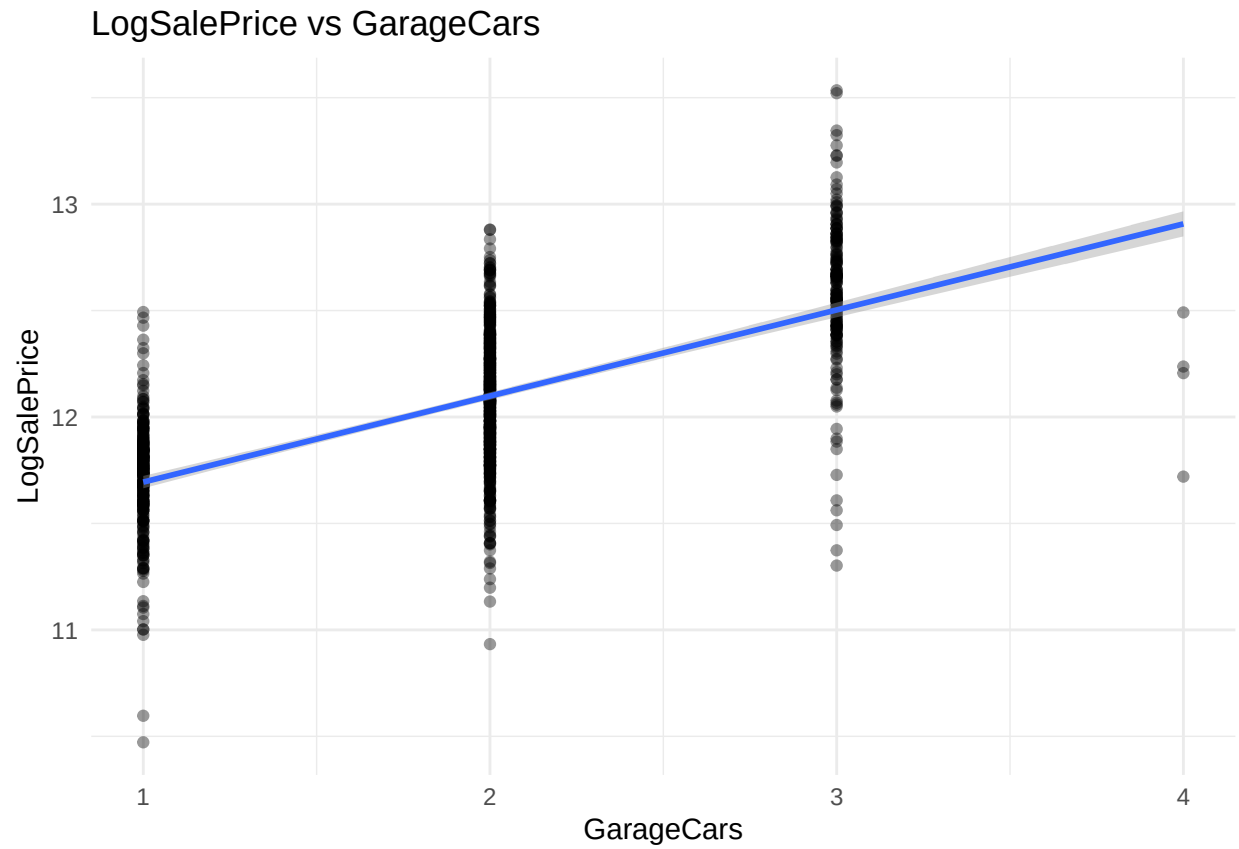
```
## `geom_smooth()` using formula = 'y ~ x'
```



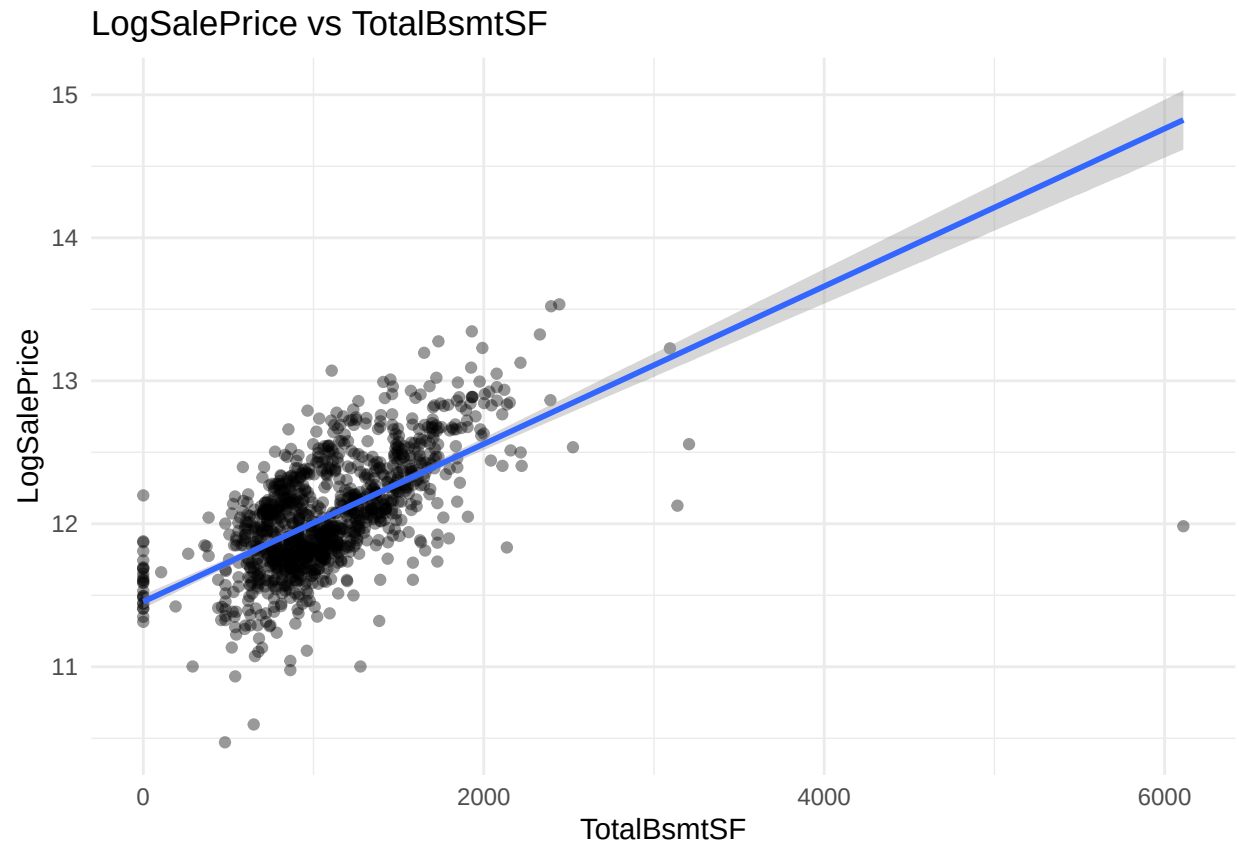
```
## `geom_smooth()` using formula = 'y ~ x'
```



```
## `geom_smooth()` using formula = 'y ~ x'
```

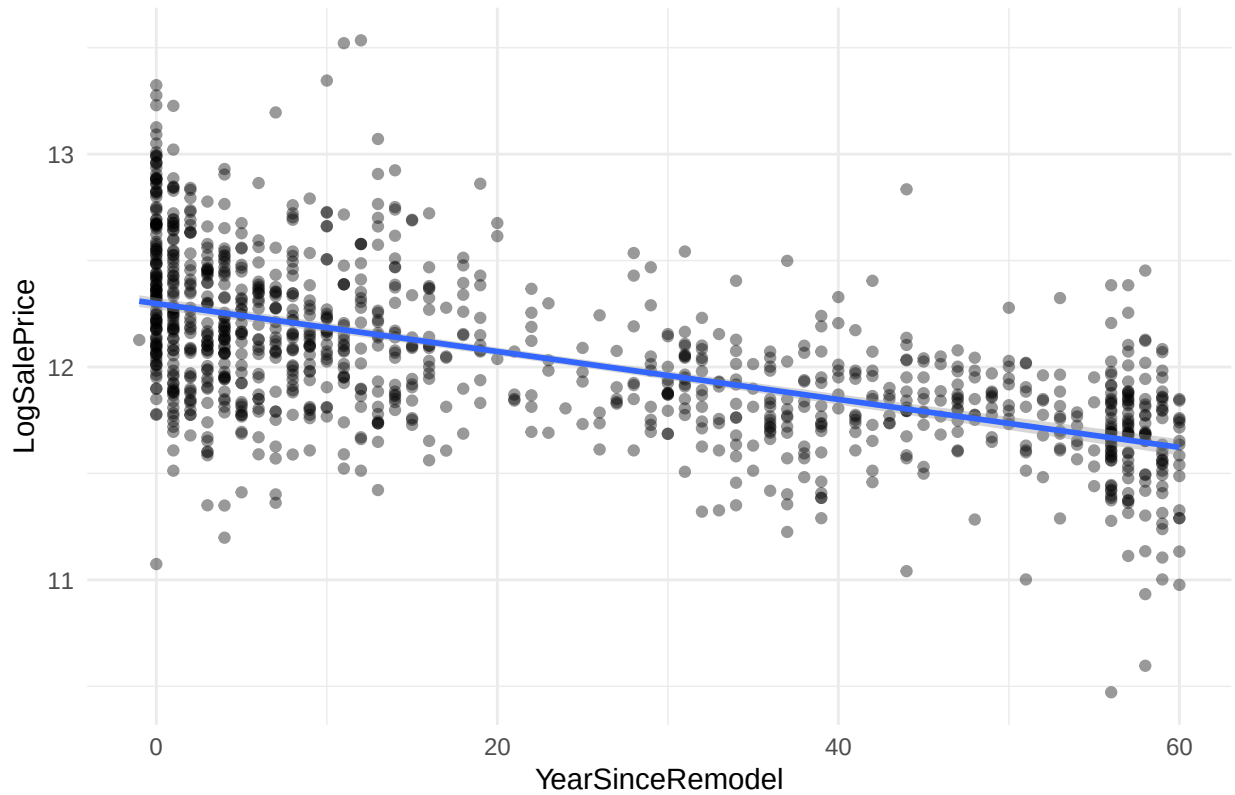


```
## `geom_smooth()` using formula = 'y ~ x'
```



```
## `geom_smooth()` using formula = 'y ~ x'
```

LogSalePrice vs YearSinceRemodel



$$y_i = \log(\text{SalePrice}_i), \quad i = 1, \dots, n$$

$$y_i \sim \mathcal{N}(\mu_i, \sigma^2)$$

$$\mu_i = \beta_0 + \alpha_{g[i]} + \beta_1 \text{OverallQual}_i + \beta_2 \text{LogGrLivArea}_i + \beta_3 \text{GarageCars}_i + \beta_4 \text{TotalBsmtSF}_i + \beta_5 \text{YearSinceRemodel}_i$$

$$\text{LogGrLivArea}_i = \log(\text{GrLivArea}_i)$$

$$\beta_0 \sim \mathcal{N}(8.8, 0.05^2), \quad \beta_1 \sim \mathcal{N}(0.10, 0.03^2), \quad \beta_2 \sim \mathcal{N}(0.25, 0.08^2),$$

$$\beta_3 \sim \mathcal{N}(0.06, 0.03^2), \quad \beta_4 \sim \mathcal{N}(0.00010, 0.00003^2), \quad \beta_5 \sim \mathcal{N}(-0.003, 0.004^2)$$

$$\alpha_j \sim \mathcal{N}(0, \tau_{\text{nbhd}}^2), \quad j = 1, \dots, J$$

$$\tau_{\text{nbhd}} \sim \text{half-Normal}(0, 0.12^2), \quad \sigma \sim \text{half-Normal}(0, 0.10^2)$$

```
# data for prior predictive check
house_price_prior <- house_price |>
  mutate(
    LogGrLivArea = log(GrLivArea),
    Neighborhood_id = as.integer(Neighborhood)
  ) |>
  select(
    LogSalePrice,
    OverallQual,
    LogGrLivArea,
    GarageCars,
    TotalBsmtSF,
    YearSinceRemodel,
```

```

    Neighborhood_id
  )

J <- length(unique(house_price_prior$Neighborhood_id))
# draw one set of priors
draw_prior_params <- function() {
  list(
    beta0 = simulate(Norm(8.8, 0.05)),
    beta1 = simulate(Norm(0.10, 0.03)),      # OverallQual
    beta2 = simulate(Norm(0.25, 0.08)),      # LogGrLivArea
    beta3 = simulate(Norm(0.06, 0.03)),      # GarageCars
    beta4 = simulate(Norm(0.00010, 0.00003)), # TotalBsmtSF
    beta5 = simulate(Norm(-0.003, 0.004)),   # YearSinceRemodel
    tau_nbhd = abs(simulate(Norm(0, 0.12))),
    sigma_y = abs(simulate(Norm(0, 0.10)))
  )
}

plot_prior_lines_one_var <- function(df, xvar, n_draws = 40) {
  xvals <- df[[xvar]]
  yvals <- df$LogSalePrice

  xs <- seq(
    min(xvals, na.rm = TRUE),
    max(xvals, na.rm = TRUE),
    length.out = 100
  )

  oq0 <- median(df$OverallQual, na.rm = TRUE)
  lg0 <- median(df$LogGrLivArea, na.rm = TRUE)
  gc0 <- median(df$GarageCars, na.rm = TRUE)
  bs0 <- median(df$TotalBsmtSF, na.rm = TRUE)
  yr0 <- median(df$YearSinceRemodel, na.rm = TRUE)

  plot(
    xvals, yvals,
    pch = 16,
    cex = 0.45,
    col = rgb(0, 0, 0, 0.10),
    xlab = xvar,
    ylab = "log(SalePrice)",
    main = paste("Prior lines for", xvar),
    ylim = c(10.5, 13.8)
  )

  for (i in 1:n_draws) {
    par_draw <- draw_prior_params()

    OverallQual_x <- rep(oq0, length(xs))
    LogGrLivArea_x <- rep(lg0, length(xs))
    GarageCars_x <- rep(gc0, length(xs))
    TotalBsmtSF_x <- rep(bs0, length(xs))
    YearSinceRemodel_x <- rep(yr0, length(xs))
  }
}

```

```

    if (xvar == "OverallQual") {
      OverallQual_x <- xs
    }
    if (xvar == "LogGrLivArea") {
      LogGrLivArea_x <- xs
    }
    if (xvar == "GarageCars") {
      GarageCars_x <- xs
    }
    if (xvar == "TotalBsmtSF") {
      TotalBsmtSF_x <- xs
    }
    if (xvar == "YearSinceRemodel") {
      YearSinceRemodel_x <- xs
    }

    # hold neighborhood effect at 0 for slope check
    mu_line <- par_draw$beta0 +
      par_draw$beta1 * OverallQual_x +
      par_draw$beta2 * LogGrLivArea_x +
      par_draw$beta3 * GarageCars_x +
      par_draw$beta4 * TotalBsmtSF_x +
      par_draw$beta5 * YearSinceRemodel_x

    lines(xs, mu_line, col = rgb(0, 0, 1, 0.18))
  }
}
set.seed(1)

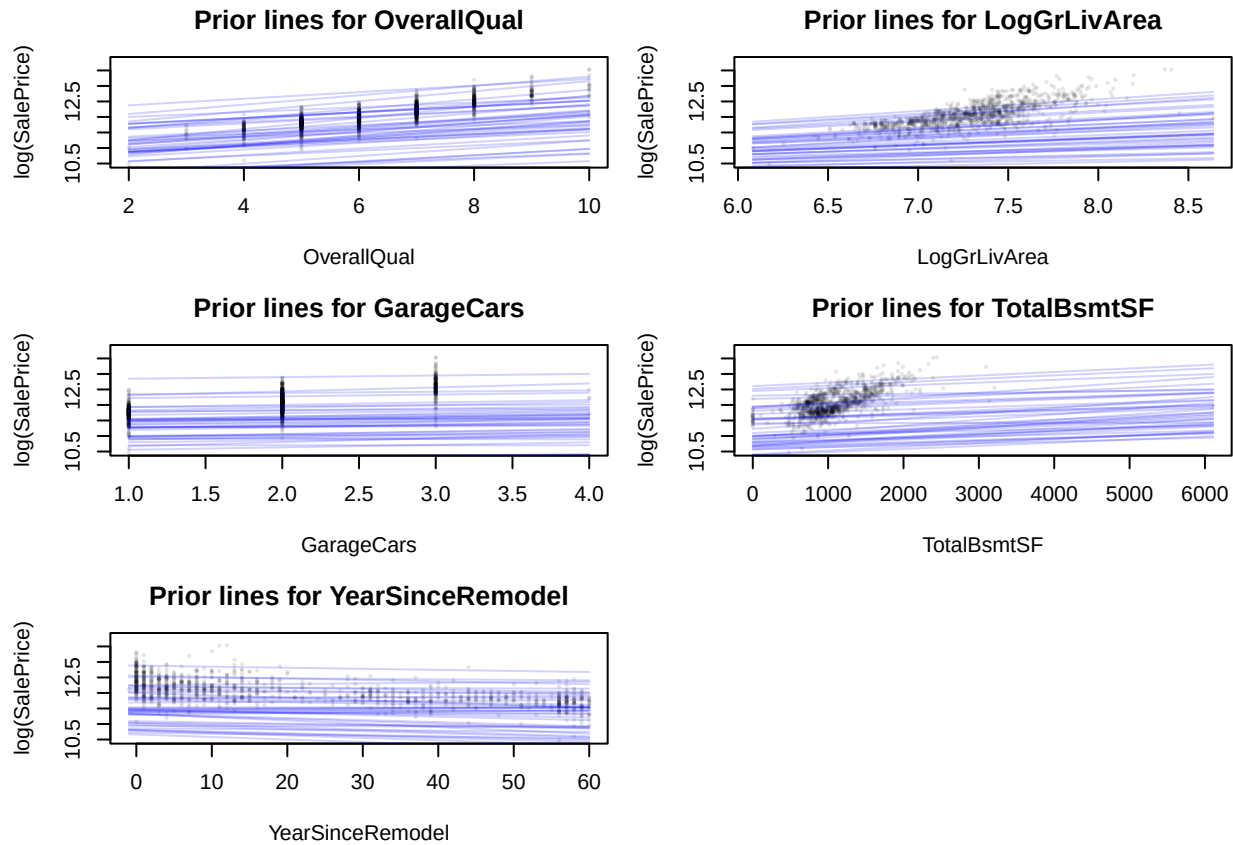
par(mfrow = c(3, 2), mar = c(4, 4, 3, 1))

vars_check <- c(
  "OverallQual",
  "LogGrLivArea",
  "GarageCars",
  "TotalBsmtSF",
  "YearSinceRemodel"
)

for (v in vars_check) {
  plot_prior_lines_one_var(house_price_prior, v, n_draws = 40)
}

plot.new()

```

```
par(mfrow = c(1, 1))

prior_rep_once <- function(df) {
  par_draw <- draw_prior_params()

  alpha <- sapply(1:J, function(j) simulate(Norm(0, par_draw$tau_nbhd)))

  mu <- par_draw$beta0 +
    par_draw$beta1 * df$OverallQual +
    par_draw$beta2 * df$LogGrLivArea +
    par_draw$beta3 * df$GarageCars +
    par_draw$beta4 * df$TotalBsmtSF +
    par_draw$beta5 * df$YearSinceRemodel +
    alpha[df$Neighborhood_id]

  yrep <- sapply(mu, function(m) simulate(Norm(m, par_draw$sigma_y)))
  return(yrep)
}

set.seed(1)

plot(
  NULL,
  xlim = c(9, 15),
  ylim = c(0, 2.5),
  xlab = "log(SalePrice)",
  ylab = "Density",
```

```

  main = "Prior predictive densities"
)

# observed density
obs_d <- density(house_price_prior$LogSalePrice)
lines(obs_d, col = "red", lwd = 3)

# prior predictive densities
for (s in 1:30) {
  yrep <- prior_rep_once(house_price_prior)
  d <- density(yrep)
  lines(d, col = rgb(0, 0, 1, 0.15))
}

legend(
  "topright",
  legend = c("Observed", "Prior predictive"),
  col = c("red", "blue"),
  lwd = c(3, 1)
)

```

